Mock Analysis — Group 1

Predicting Blood Pressure from NHANES

The Data

The National Health and Nutrition Examination Survey (NHANES) is a large, ongoing, nationally representative health survey run by the CDC. NHANES uses a complex survey design where participants are sampled within primary sampling units (`SDMVPSU`) nested in strata (`SDMVSTRA`), and observations within the same PSU are correlated. You will pull one or more recent NHANES cycles — the demographics file, the blood pressure exam file, and the body measures file — and link them by participant ID to build an analysis dataset.

Your predictors can include age, sex, BMI, and any other participant characteristics you think are relevant from the files you pull. Take a look at what else is included in the demographics file beyond the variables you'd use as predictors — not everything in there is meant to go into your model as a predictor.

Your Goal

Build a model that predicts systolic blood pressure from participant characteristics, and give an honest estimate of how well it will perform on a new NHANES participant.

Task 1 — Designing the Cross-Validation Procedure (≈ 10 min)

As a group, discuss what you would be most careful about when designing the cross-validation procedure for this analysis. Consider, among other things:

- Temporal ordering
- Clustering (observations that are not independent of one another)
- Calculating error by subgroup
- Data preprocessing and leakage
- Mismatch between your training data and the population/context you actually want to predict for

1. What is the main CV issue you are concerned about here, and why?
2. How would you design the CV procedure to address it? Be specific: what defines a fold, what gets held out together, and why.

Task 2 — Choosing an Analysis Approach (≈ 10 min)

So far this semester we have covered the following methods:

- Linear regression
- Ridge regression
- Lasso
- Group lasso
- Fused lasso
- Generalized additive models (GAMs)

1. Which of these methods fits this analysis best, and why?
2. What is it about the structure of the predictors, or the goal of the analysis, that rules out (or supports) each option?

Be ready to present: your CV design, your chosen method, and your reasoning — 3–4 minutes per group.

Mock Analysis — Group 2

Forecasting Flu Testing Volume and Percent Positive

The Data

CDC's clinical laboratory surveillance system (ILINet/NREVSS) reports weekly counts of influenza specimens tested and percent positive, by state. You have this data for the five most populous states, spanning several flu seasons across multiple years.

Your Goal

Predict total weekly specimens tested and percent positive for a specific state in a future week — for example, California in week 50 of 2026 — and report your expected forecast error for that prediction.

Task 1 — Designing the Cross-Validation Procedure (≈ 10 min)

As a group, discuss what you would be most careful about when designing the cross-validation procedure for this analysis. Consider, among other things:

- Temporal ordering
- Clustering (observations that are not independent of one another)
- Calculating error by subgroup
- Data preprocessing and leakage
- Mismatch between your training data and the population/context you actually want to predict for

1. What is the main CV issue you are concerned about here, and why?
2. How would you design the CV procedure to address it? Be specific: what defines a fold, what gets held out together, and why.

Task 2 — Choosing an Analysis Approach (≈ 10 min)

So far this semester we have covered the following methods:

- Linear regression
- Ridge regression
- Lasso
- Group lasso
- Fused lasso
- Generalized additive models (GAMs)

1. Which of these methods fits this analysis best, and why?
2. What is it about the structure of the predictors, or the goal of the analysis, that rules out (or supports) each option?

Be ready to present: your CV design, your chosen method, and your reasoning — 3–4 minutes per group.

Mock Analysis — Group 3

Predicting Wheat Yield from Genetic Markers

The Data

You have a panel of wheat breeding lines, each genotyped at over a thousand genetic markers. Yield has been measured for each line in each of several distinct growing environments. There are 4 environments in total, each uses the same lines, but has a different environment/conditions.

Your Goal

Build a model that predicts yield from the genetic markers, training it using data from some of the environments, and evaluate how well it predicts yield in an environment that was not used for training.

Task 1 — Designing the Cross-Validation Procedure (≈ 10 min)

As a group, discuss what you would be most careful about when designing the cross-validation procedure for this analysis. Consider, among other things:

- Temporal ordering
- Clustering (observations that are not independent of one another)
- Calculating error by subgroup
- Data preprocessing and leakage
- Mismatch between your training data and the population/context you actually want to predict for

1. What is the main CV issue you are concerned about here, and why?
2. How would you design the CV procedure to address it? Be specific: what defines a fold, what gets held out together, and why.

Task 2 — Choosing an Analysis Approach (≈ 10 min)

So far this semester we have covered the following methods:

- Linear regression
- Ridge regression
- Lasso
- Group lasso
- Fused lasso
- Generalized additive models (GAMs)

1. Which of these methods fits this analysis best, and why?
2. What is it about the structure of the predictors, or the goal of the analysis, that rules out (or supports) each option?

Be ready to present: your CV design, your chosen method, and your reasoning — 3–4 minutes per group.